Formal Verification of BLR PCP

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0.1 Preliminaries

Let $q = p^s$ for some prime p and $s \in \mathbb{N}$. Let $f : \mathbb{F}^n \to \mathbb{F}_q$ be a function over the finite field $\mathbb{F}_q$. Let $\omega_p = e^{2\pi i/p}$ be the root of unity modolu p.

Definition 0.1.1 (Trace function in finite fields)**.** *The trace function* $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ *is defined as follows:*

$$\mathrm{Tr}(x) = \sum_{i=0}^{s-1} x^{p^i}$$

Definition 0.1.2. *The character group of* $\mathbb{F}_q$*, denoted* $\hat{G}$ *, contains all homomorphisms* $\chi : \mathbb{F}^n \to \mathbb{C}$.

Fact 0.1.3. $\hat{G}$ *consists of all the functions:* $\chi_\alpha(x) = \omega_p^{\mathrm{Tr}(\langle \alpha, x \rangle)}$*, for all* $\alpha \in \mathbb{F}^n$.

Fact 0.1.4. $\{\chi_\alpha\}_{\alpha \in \mathbb{F}^n}$ *form an orthonormal basis for all functions* $f : \mathbb{F}_q^n \to \mathbb{C}$ *under the inner-product* $\langle f, g \rangle = \mathbb{E}_{x \in \mathbb{F}^n}[f(x)\overline{g(x)}]$.

Definition 0.1.5. *Every function* $h : \mathbb{F}^n \to \mathbb{C}$ *has a Fourier expansion* $\hat{h} : \mathbb{F}^n \to \mathbb{C}$ *given by*

$$h(x) = \sum_{\alpha \in \mathbb{F}_q} \hat{h}(\alpha)\chi_\alpha(x)$$

$$\hat{h}(\alpha) = q^{-n} \sum_{x \in \mathbb{F}_q} h(x)\overline{\chi_\alpha(x)}$$

Definition 0.1.6. *The norm of a function* $h : \mathbb{F}^n \to \mathbb{C}$ *is equal to:*

$$\|h\|^2 = \langle h, h \rangle = \mathbb{E}_{x \in \mathbb{F}^n}[h(x)\overline{h(x)}] = q^{-n} \sum_{x \in \mathbb{F}^n} h(x)^2$$

Corollary 0.1.7. *Let* $h : \mathbb{F}^n \to \mathbb{C}$*. Then:* $\|h\| \leq 1$.

Proof.

$$\|h\|^2 = q^{-n} \sum_{x \in \mathbb{F}^n} h(x)^2 \leq q^{-n} \sum_{x \in \mathbb{F}^n} |h(x)|^2 \leq 1$$

□

Theorem 0.1.8. *Let* $h_1, h_2 : \mathbb{F}^n \to \mathbb{C}$.

- *Parseval's Identity:* $\|h_1\|^2 = \sum_{\alpha \in \mathbb{F}^n} \widehat{h_1}(\alpha)^2$.
- *Plancheral's Identity:* $\langle h_1, h_2 \rangle = \sum_{\alpha \in \mathbb{F}^n} \widehat{h_1}(\alpha)\widehat{h_2}(\alpha)$.

As there is no isomorphism between $\mathbb{F}_q$ and $\mathbb{C}$, we instead consider $q-1$ functions $\varphi_c : \mathbb{F}^n \to \mathbb{C}$ for all $c \in \mathbb{F}_q^*$:

$$\varphi_c(x) := \omega_p^{\mathrm{Tr}(c \cdot f(x))} = \sum_{\alpha \in \mathbb{F}^n} \widehat{(\varphi_c)}(\alpha)\chi_\alpha(x) \tag{1}$$

Definition 0.1.9. *For any* $f : \mathbb{F}^n \to \mathbb{F}_q$ *we defined the* $\mathbb{F}_p$*-expansion of* f *as the set:* $\{\varphi_c\}_{c \in \mathbb{F}_q^*}$.

Definition 0.1.10. *Let $f, g : \mathbb{F}^n \to \mathbb{F}_q$. We defined the distance between f, g as:*

$$\delta(f, g) = \Pr_{x \in \mathbb{F}^n} [f(x) \neq g(x)]$$

Now, we express the distance of f, g in terms of the Fourier coefficients $\mathbb{F}_p$-expansion sets.

Claim 0.1.11. *Let $f, g : \mathbb{F}^n \to \mathbb{F}_q$ and their $\mathbb{F}_p$-expansion sets be $\{\varphi_c\}_{c \in \mathbb{F}_q^*}, \{\gamma_c\}_{c \in \mathbb{F}_q^*}$ respectively.*

$$\delta(f, g) = 1 - \frac{1}{q}\left(1 + \sum_{c \in \mathbb{F}_q^*} \langle \varphi_c, \gamma_c \rangle\right) = 1 - \frac{1}{q}\left(1 + \sum_{c \in \mathbb{F}_q^*} \sum_{\alpha \in \mathbb{F}^n} \widehat{\varphi_c}(\alpha)\overline{\widehat{\gamma_c}(\alpha)}\right)$$

Definition 0.1.12. *Let $\mathcal{LIN}$ be the set of linear functions form $\mathbb{F}^n$ to $\mathbb{F}_q$. For $\alpha \in \mathbb{F}^n$, we denote $\ell_\alpha(x) = \langle x, \alpha \rangle$. Then:*

$$\mathcal{LIN} = \{\ell_\alpha\}_{\alpha \in \mathbb{F}^n}$$

Claim 0.1.13. *Let $f : \mathbb{F}^n \to \mathbb{F}_q$ with $\mathbb{F}_p$-expansion sets be $\{\varphi_c\}_{c \in \mathbb{F}_q^*}$ and let $\ell_\alpha \in \mathcal{LIN}$ for some $\alpha \in \mathbb{F}^n$, and let it's $\mathbb{F}_p$-expansion set be $\{\lambda_c\}_{c \in \mathbb{F}_q^*}$. Then:*

$$\delta(f, \ell_\alpha) = 1 - \left(\frac{1}{q}\left(1 + \sum_{c \in \mathbb{F}_q^*} \widehat{\varphi_c}(c\alpha)\right)\right)$$

Proof. For all $\beta \in \mathbb{F}^n$, the following is 0 if $\beta \neq c\alpha$ and 1 otherwise:

$$\widehat{\lambda_c}(\beta) = q^{-n} \sum_{x \in \mathbb{F}^n} \lambda_c(x)\overline{\chi_\beta(x)} = q^{-n} \sum_{x \in \mathbb{F}^n} \omega_p^{\mathrm{Tr}(c \cdot \ell_\alpha(x)) - \mathrm{Tr}(\langle \beta, x \rangle)} = q^{-n} \sum_{x \in \mathbb{F}^n} \omega_p^{\mathrm{Tr}(\langle c\alpha, x \rangle) - \mathrm{Tr}(\langle \beta, x \rangle)}$$

□

Corollary 0.1.14. *For all $\alpha \in \mathbb{F}^n$, $-1 \leq \sum_{c \in \mathbb{F}_q^*} \widehat{\varphi_c}(c\alpha) \leq q - 1$ is a real number.*

Proof. This follows because $\delta(f, \ell_\alpha) \in [0, 1]$. □

Proof. Equivalent to $0 \leq \delta \leq 1 - 1/q$ □

Definition 0.1.15. *The distance of $f : \mathbb{F}^n \to \mathbb{F}_q$ from $\mathcal{LIN}$ is defined as: $\delta(f, \mathcal{LIN}) := \min_{\alpha \in \mathbb{F}^n} \delta(f, \ell_\alpha)$.*

Corollary 0.1.16.

$$\delta(f, \mathcal{LIN}) = \min_{\alpha \in \mathbb{F}^n}\left(1 - \left(\frac{1}{q}\left(1 + \sum_{c \in \mathbb{F}_q^*} \widehat{\varphi_c}(c\alpha)\right)\right)\right)$$

Equivalently,

$$\delta(f, \mathcal{LIN}) = 1 - \frac{1}{q}\left(1 + \max_{\alpha \in \mathbb{F}^n}\left(\sum_{c \in \mathbb{F}_q^*} \widehat{\varphi_c}(c\alpha)\right)\right)$$

0.2 BLR Test over Finite Fields

The BLR Test Given oracle access to a function $f : \mathbb{F}^n \to \mathbb{F}_q$, the verifier performs the following:

- Sample random $a, b \in \mathbb{F}^n \setminus \{0\}$ and sample $x, y \in \mathbb{F}^n$.
- $\mathsf{V}^f_{\mathsf{BLR}} = 1 \iff a \cdot f(x) + b \cdot f(y) = f(a \cdot x + b \cdot y)$.

Theorem 0.2.1. *Let $f : \mathbb{F}^n \to \mathbb{F}_q$ with $\mathbb{F}_p$-expansion sets be $\{\varphi_c\}_{c \in \mathbb{F}_q^*}$.*

$$\Pr_{a,b,x,y}\left[\mathsf{V}^f_{\mathsf{BLR}} = 1\right] = \frac{1}{q}\left(1 + \frac{1}{(q-1)^2}\sum_{\alpha \in \mathbb{F}^n}\left(\sum_{a \in \mathbb{F}_q^*} \widehat{\varphi_a}(a\alpha)\right)^3\right)$$

Proof. Let the $\mathbb{F}_p$-expansion set of f be $\{\varphi_c\}_{c \in \mathbb{F}_q^*}$. Let also $a, b \in \mathbb{F}^n \setminus \{0\}$ and $x, y \in \mathbb{F}^n$. We observe that:

$$\mathsf{V}^f_{\mathsf{BLR}} = 1 \iff \frac{1}{q}\left(\sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(af(x)+bf(y)-f(ax+by)}\right) = 1$$

As the above sum is zero unless $af(x) + bf(y) - f(ax + by) = 0$. Hence:

$$\begin{aligned}\Pr_{a,b,x,y}\left[\mathsf{V}^f_{\mathsf{BLR}} = 1\right] &= \mathbb{E}_{a,b,x,y}\left[\frac{1}{q}\left(\sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(af(x)+bf(y)-f(ax+by))}\right)\right] \\ &= \frac{1}{q} + \mathbb{E}_{a,b,x,y}\left[\sum_{c \in \mathbb{F}_q^*} \omega_p^{\mathrm{Tr}(c(af(x)+bf(y)-f(ax+by))}\right]\end{aligned}$$

Now, we consider the last term:

$$\begin{aligned}&\mathbb{E}_{a,b,x,y}\left[\sum_{c \in \mathbb{F}_q^*} \omega_p^{\mathrm{Tr}(c(af(x)+bf(y)-f(ax+by))}\right] \\ &= \mathbb{E}_{a,b,x,y}\left[\sum_{c \in \mathbb{F}_q^*} \varphi_{ca}(x)\varphi_{cb}(y)\varphi_{-c}(a \cdot x + b \cdot y)\right] \\ &= \mathbb{E}_{a,b,x,y}\left[\sum_{c \in \mathbb{F}_q^*}\sum_{\alpha,\beta,\gamma \in \mathbb{F}^n} \widehat{\varphi_{ca}}(\alpha)\chi_\alpha(x)\widehat{\varphi_{cb}}(\beta)\chi_\beta(y)\widehat{\varphi_{-c}}(\gamma)\chi_\gamma(ax+by)\right] \\ &= \mathbb{E}_{a,b,x,y}\left[\sum_{c \in \mathbb{F}_q^*}\sum_{\alpha,\beta,\gamma \in \mathbb{F}^n} \widehat{\varphi_{ca}}(\alpha)\widehat{\varphi_{cb}}(\beta)\widehat{\varphi_{-c}}(\gamma)\chi_\alpha(x)\chi_\beta(y)\chi_\gamma(ax+by)\right] \\ &= \mathbb{E}_{a,b,x,y}\left[\sum_{c \in \mathbb{F}_q^*}\sum_{\alpha,\beta,\gamma \in \mathbb{F}^n} \widehat{\varphi_{ca}}(\alpha)\widehat{\varphi_{cb}}(\beta)\widehat{\varphi_{-c}}(\gamma)\chi_\alpha(x)\chi_\beta(y)\chi_{a\gamma}(x)\chi_{b\gamma}(y)\right] \\ &= \mathbb{E}_{a,b}\left[\sum_{c \in \mathbb{F}_q^*}\sum_{\alpha,\beta,\gamma \in \mathbb{F}^n} \widehat{\varphi_{ca}}(\alpha)\widehat{\varphi_{cb}}(\beta)\widehat{\varphi_{-c}}(\gamma)\mathbb{E}_{x,y}\left[\chi_\alpha(x)\chi_\beta(y)\chi_{a\gamma}(x)\chi_{b\gamma}(y)\right]\right]\end{aligned}$$

Now, we consider the following expression:

$$\mathbb{E}_{x,y}\left[\chi_\alpha(x)\chi_\beta(y)\chi_{a\gamma}(x)\chi_{b\gamma}(y)\right] = \mathbb{E}_{x,y}\left[\omega_p^{\mathrm{Tr}(\langle\alpha,x\rangle+\langle\beta,y\rangle+\langle\gamma,ax\rangle+\langle\gamma,by\rangle)}\right]$$

Observe that the above is zero, unless $\gamma = -a^{-1}\alpha$ and $\beta = a^{-1}b\alpha$. Thus:

$$\begin{aligned}
&\mathbb{E}_{a,b,x,y}\left[\sum_{c\in\mathbb{F}_q^*}\omega_p^{\mathrm{Tr}(c(af(x)+bf(y)-f(ax+by))}\right]\\
&= \mathbb{E}_{a,b}\left[\sum_{c\in\mathbb{F}_q^*}\sum_{\alpha,\beta,\gamma\in\mathbb{F}^n}\widehat{\varphi_{ca}}(\alpha)\widehat{\varphi_{cb}}(\beta)\widehat{\varphi_{-c}}(\gamma)\mathbb{E}_{x,y}\left[\chi_\alpha(x)\chi_\beta(y)\chi_{a\gamma}(x)\chi_{b\gamma}(y)\right]\right]\\
&= \mathbb{E}_{a,b}\left[\sum_{c\in\mathbb{F}_q^*}\sum_{\alpha\in\mathbb{F}^n}\widehat{\varphi_{ca}}(\alpha)\widehat{\varphi_{cb}}(a^{-1}b\alpha)\widehat{\varphi_{-c}}(-a^{-1}\alpha)\right]\\
&= \mathbb{E}_{a,b}\left[\sum_{c\in\mathbb{F}_q^*}\sum_{\alpha\in\mathbb{F}^n}\widehat{\varphi_{ca}}(ca\alpha)\widehat{\varphi_{cb}}(cb\alpha)\widehat{\varphi_{-c}}(-c\alpha)\right]\\
&= \mathbb{E}_{a,b}\left[\sum_{c\in\mathbb{F}_q^*}\sum_{\alpha\in\mathbb{F}^n}\widehat{\varphi_{a}}(a\alpha)\widehat{\varphi_{b}}(b\alpha)\widehat{\varphi_{c}}(c\alpha)\right]\\
&= \frac{1}{(q-1)^2}\sum_{a,b,c\in\mathbb{F}_q^*}\sum_{\alpha\in\mathbb{F}^n}\widehat{\varphi_{a}}(a\alpha)\widehat{\varphi_{b}}(b\alpha)\widehat{\varphi_{c}}(c\alpha)\\
&= \frac{1}{(q-1)^2}\sum_{\alpha\in\mathbb{F}^n}\left(\sum_{a\in\mathbb{F}_q^*}\widehat{\varphi_{a}}(a\alpha)\right)^3
\end{aligned}$$

□

Theorem 0.2.2. *Let $f:\mathbb{F}^n\to\mathbb{F}_q$.*

$$(1-\delta(f,\mathcal{LIN}))^3 \le \Pr_{a,b,x,y}\left[\mathsf{V}^f_{\mathsf{BLR}} = 1\right] \le 1-\delta(f,\mathcal{LIN})$$

Proof. For the left hand-side:

$$\begin{aligned}
\Pr\left[\mathsf{V}^f_{\mathsf{BLR}} = 1\right] &= \frac{1}{q}\left(1+\frac{1}{(q-1)^2}\sum_{\alpha\in\mathbb{F}^n}\left(\sum_{a\in\mathbb{F}_q^*}\widehat{\varphi_a}(a\alpha)\right)^3\right)\\
&\le \frac{1}{q}\left(1+\frac{1}{(q-1)^2}\max_{\alpha\in\mathbb{F}^n}\left(\sum_{a\in\mathbb{F}_q^*}\widehat{\varphi_a}(a\alpha)\right)\sum_{\alpha\in\mathbb{F}^n}\left(\sum_{a\in\mathbb{F}_q^*}\widehat{\varphi_a}(a\alpha)\right)^2\right)\\
&= \frac{1}{q}\left(1+\frac{1}{(q-1)^2}\max_{\alpha\in\mathbb{F}^n}\left(\sum_{a\in\mathbb{F}_q^*}\widehat{\varphi_a}(a\alpha)\right)\sum_{\alpha\in\mathbb{F}^n}\left(\sum_{a,b\in\mathbb{F}_q^*}\widehat{\varphi_a}(a\alpha)\widehat{\varphi_b}(b\alpha)\right)\right)\\
&= \frac{1}{q}\left(1+\frac{1}{(q-1)^2}\max_{\alpha\in\mathbb{F}^n}\left(\sum_{a\in\mathbb{F}_q^*}\widehat{\varphi_a}(a\alpha)\right)\sum_{a,b\in\mathbb{F}_q^*}\left(\sum_{\alpha\in\mathbb{F}^n}\widehat{\varphi_a}(a\alpha)\widehat{\varphi_b}(b\alpha)\right)\right)
\end{aligned}$$

Consider the following, due to Plancherel's identity and the fact that φ_a map to the roots of unity modulo p:

$$\sum_{\alpha \in \mathbb{F}^n} \widehat{\varphi_a}(a\alpha)\widehat{\varphi_b}(b\alpha) = \mathbb{E}_\alpha \left[\varphi_a(a\alpha)\gamma_b(b\alpha)\right] \leq \mathbb{E}_\alpha \left[|\varphi_a(a\alpha)\gamma_b(b\alpha)|\right] \leq 1$$

Hence, it follows that:

$$\Pr_{a,b,x,y}\left[\mathsf{V}^f_{\mathsf{BLR}} = 1\right] \leq \frac{1}{q}\left(1 + \max_{\alpha \in \mathbb{F}^n}\left(\sum_{a \in \mathbb{F}_q^*} \widehat{\varphi_a}(a\alpha)\right)\right) = 1 - \delta(f, \mathcal{LIN})$$

For the left hand side of the claim:

$$\begin{aligned}
\Pr\left[\mathsf{V}^f_{\mathsf{BLR}} = 1\right] &= \frac{1}{q}\left(1 + \frac{1}{(q-1)^2}\sum_{\alpha \in \mathbb{F}^n}\left(\sum_{a \in \mathbb{F}_q^*} \widehat{\varphi_a}(a\alpha)\right)^3\right) \\
&\geq \frac{1}{q}\left(1 + \frac{1}{(q-1)^2}\max_{\alpha \in \mathbb{F}^n}\left(\sum_{a \in \mathbb{F}_q^*} \widehat{\varphi_a}(a\alpha)\right)^3\right) \\
&\geq \left(\frac{1}{q}\left(1 + \max_{\alpha \in \mathbb{F}^n}\left(\sum_{a \in \mathbb{F}_q^*} \widehat{\varphi_a}(a\alpha)\right)\right)\right)^3 \\
&= \left(1 - \delta(f, \mathcal{LIN})\right)^3
\end{aligned}$$

where the last inequality follows from the following: for every $q \geq 2, A \geq \frac{1-|\mathbb{F}|^2}{2q-1}$:

$$\left(\frac{1}{q}(1+A)\right)^3 \leq \frac{1}{q}\left(1 + \frac{1}{(q-1)^2}A^3\right)$$

(we plug $A = \max_{\alpha \in \mathbb{F}^n}\left(\sum_{a \in \mathbb{F}_q^*} \widehat{\varphi_a}(a\alpha)\right)$ in the expression above. By **??**, we observe that the above holds for any q).

□

Chapter 1

Gowers Hatami theorem

Chapter 2

Linear PCP for linear equations

Definition 2.0.1. *A system of m quadratic equations in n variables over a field $\mathbb{F}$ is a list of polynomials $p_1, \ldots, p_m \in \mathbb{F}[x_1, \ldots, x_n]$ where each p_i has total degree at most 2.*

$\mathrm{QESAT}(\mathbb{F}) := \{(p_1, \ldots, p_m) \mid \ a_1, \ldots, a_n \mathbb{F},\, i[m],\, p_i(a_1, \ldots, a_n) = 0\}$.

Definition 2.0.2. *A PCP prover is a function that takes an input x and produces a proof $\pi : []ß\Sigma$, represented as a function from an index set to the PCP alphabet. is the proof length.*

Definition 2.0.3. *A PCP verifier is an oracle machine that takes an input x and has access to two oracles: a randomness oracle, and a proof oracle.*

Definition 2.0.4. *A language $L \subseteq \{0,1\}^*$ is in $\mathrm{PCP}[_{c,s}, , , q, r]$ if there exists a PCP prover P, a PCP verifier V and a polynomial t such that for every $x\{0,1\}^*$ of size n, P produces a proof π of length $\ell(n)$, V runs in time at most $t(n)$, makes at most $q(n)$ queries to π and uses at most $r(n)$ bits of randomness, and the following holds:*

- *Completeness: If $x \in L$, then $\Pr\left[V^{\pi}(x) = 1 \mid \pi \leftarrow P(x)\right] \geq 1 - \varepsilon_c$.*
- *Soundness: If $x \notin L$, then $\forall \widetilde{\pi}$, $\Pr\left[V^{\widetilde{\pi}}(x) = 1\right] \leq \varepsilon_s$.*

Theorem 2.0.5. $\mathrm{QESAT}(\mathbb{F}_2)\mathrm{PCP}[_c = 0, _s = 1/2, \Sigma = \mathbb{F}_2, = \exp(n), q = O(1), r = n^{O(1)}]$.

Proof. □

Chapter 3

Linear PCP for tensor checks

Chapter 4

Linear PCP to PCP